

# LAKSHYA

## JEE 2027

Lecture No.- 25

**MATHEMATICS**

**RELATIONS & FUNCTIONS**

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# Topics *to be covered*

**1 Properties of Inverse of a Function**

**2 Even & Odd Functions**

**3 Question Practice**

**4**



# Relation & Function



## Relations ✓

- ★ (i) \*\* Types of Relations ✓
- ★ (ii) Number of Relations ✓
- ★ (iii) Case Making Questions ✓

## Functions

- (i) Domain
- (ii) Types of Functions
- ★ (iii) Questions on GIF & FPF,  $\text{sgn}(x)$
- (iv) Method to find Range
- (v) Equal & Identical Function
- (vi) Classification of Functions ★★
- ★ ★ ★ (vii) Number of Functions (P & C based Questions)
- (viii) Composite Function
- (ix) Inverse of Function
- (x) Even & Odd Function
- (xi) Periodic Functions
- (xii) Functional Equal
- (xiii) Graphs (Transformation & Questions)

7 Method.  
=

# P & C Based

non-uniform  
wala part





# Question Practice

## Question [JEE Mains-2022]

Let  $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ . Define  $f: S \rightarrow S$  as

$$f(n) = \begin{cases} 2n, & \text{if } n = 1, 2, 3, 4, 5 \\ 2n - 11, & \text{if } n = 6, 7, 8, 9, 10 \end{cases}$$

Let  $g: S \rightarrow S$  be a function such that  $f \circ g(n) = \begin{cases} n + 1, & \text{if } n \text{ is odd} \\ n - 1, & \text{if } n \text{ is even} \end{cases}$ ,

then  $g(10)((g(1) + g(2) + g(3) + g(4) + g(5)))$  is equal to

Ans.

## Question [JEE Mains-2021]



Let  $f, g: \mathbb{N} \rightarrow \mathbb{N}$  such that  $f(n+1) = f(n) + f(1) \forall n \in \mathbb{N}$  and  $g$  be any arbitrary function. Which of the following statements is NOT true?

- A** If  $f \circ g$  is one - one, then  $g$  is one - one
- B**  $f$  is one-one
- C** If  $g$  is onto, then  $f \circ g$  is one-one
- D** If  $f$  is onto, then  $f(n) = n \forall n \in \mathbb{N}$

Ans.





## LAST CLASS REVISION

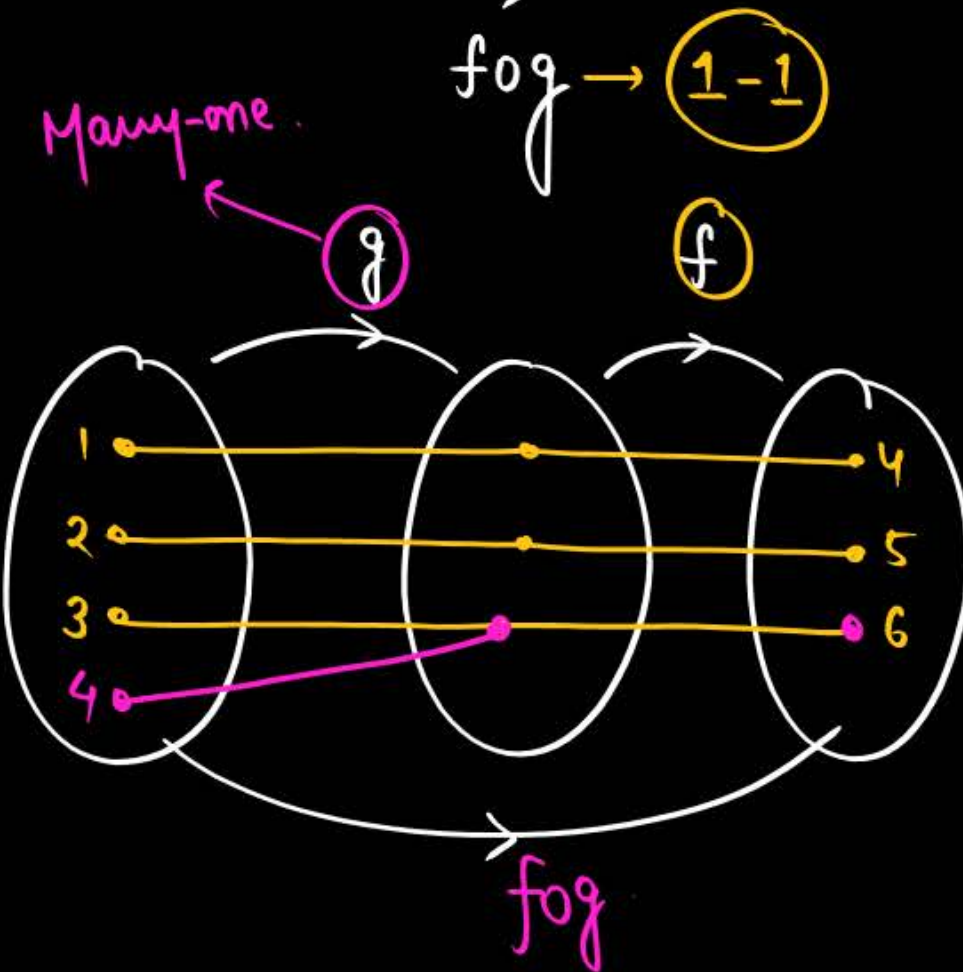
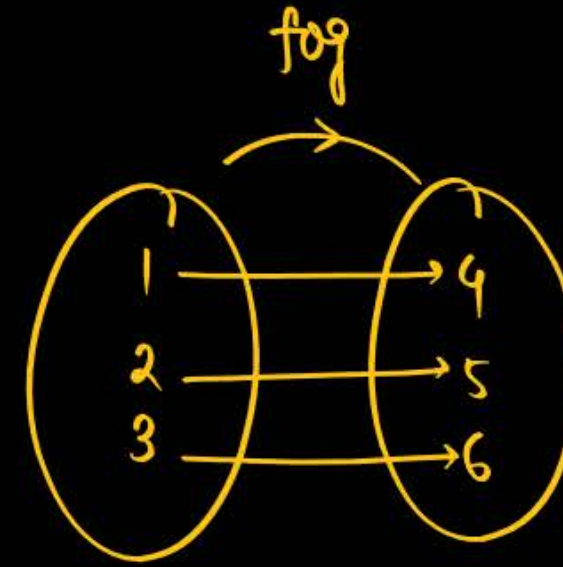
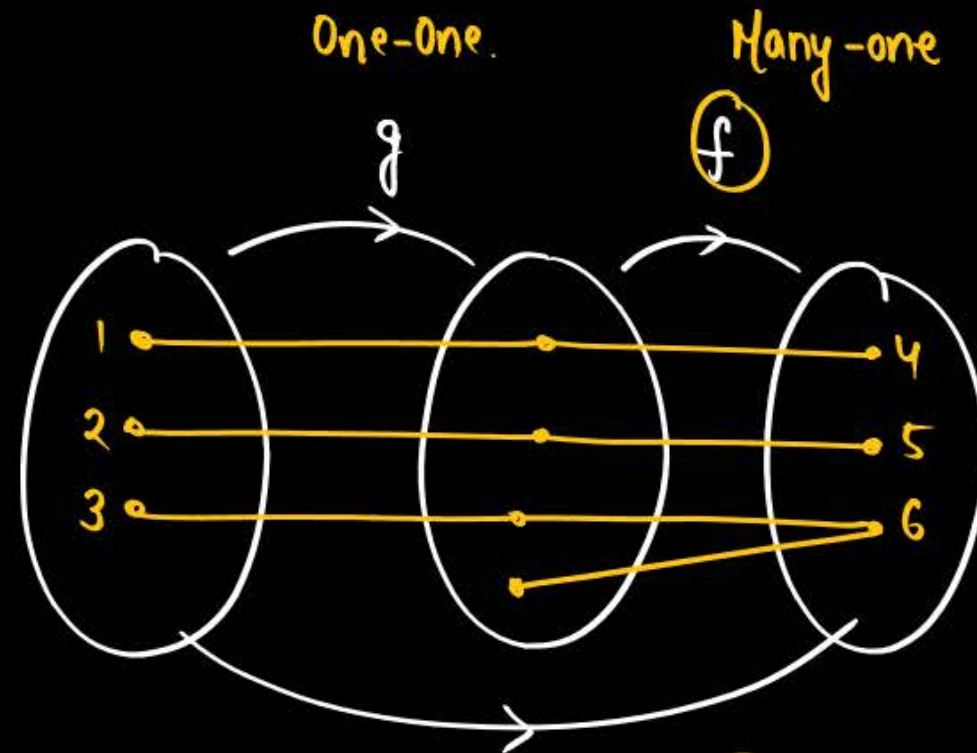


# Answer the following Questions:

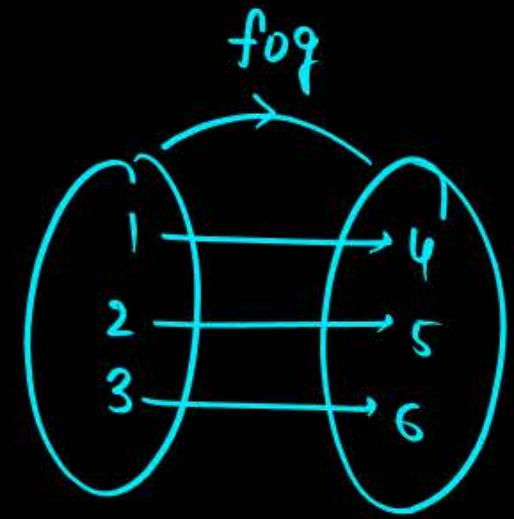
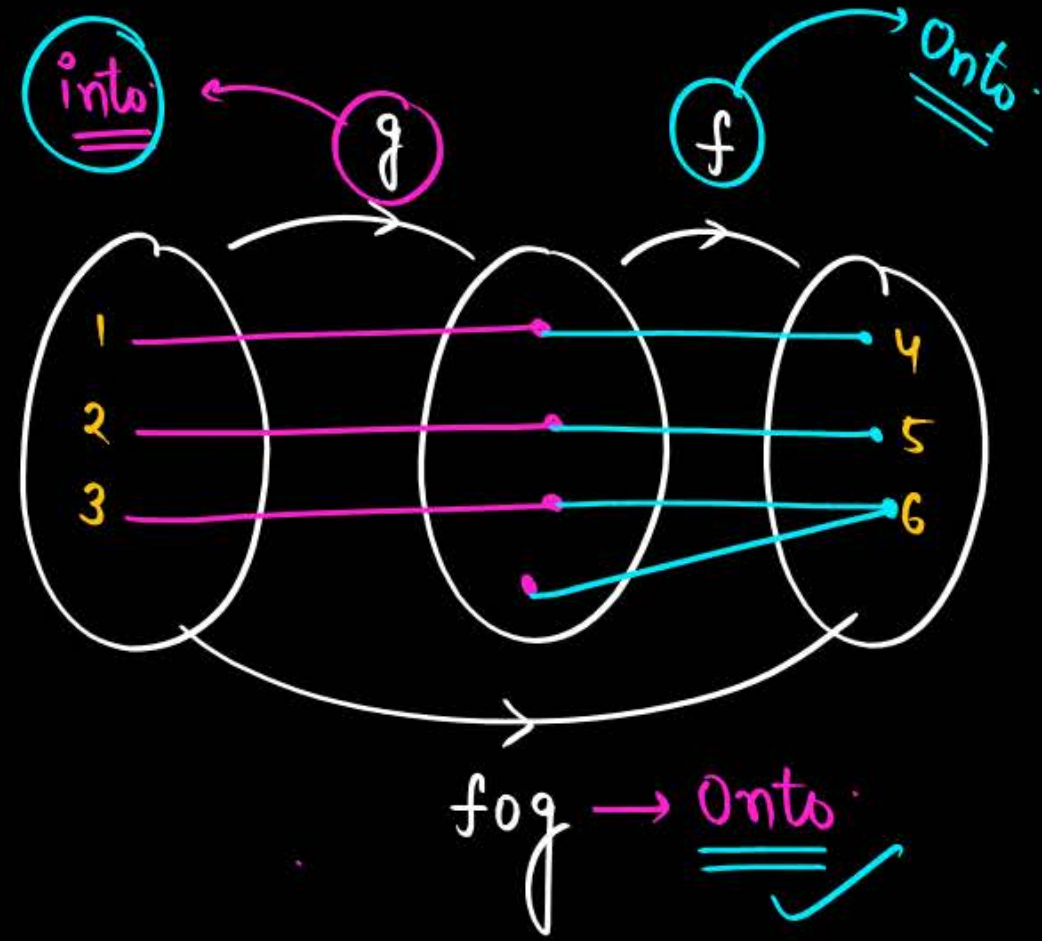
1. If  $f$  &  $g$  both are one-one then  $f \circ g$  &  $g \circ f$  should be one-one. (T)
2. If  $f$  &  $g$  both are onto then  $f \circ g$  &  $g \circ f$  should be onto. (T)
3. If  $f \circ g$  is one-one then  $g$  must be one-one where as  $f$  may or may not one-one
4. If  $f \circ g$  is onto then  $f$  must be onto where as  $g$  may or may not onto. (T)
5. If  $f \circ g$  is bijective then  $g(x)$  is 1-1 &  $f(x)$  is Onto.

1-1 & Onto

$f(g(x))$









## HOW TO FIND INVERSE??

**S-I:** Check given function is **BIJECTIVE** or not.

**S-II:** Write  $y = f(x)$  & interchange  $x$  &  $y$ .

**S-III:** Now obtain 'y' alone in terms of 'x'.

**S-IV:** In case of multiple answer, use point verification method.

$$\begin{array}{l} \checkmark (x_1, y_1) \text{ on } f(x) \checkmark \\ (y_1, x_1) \text{ on } f^{-1}(x) \checkmark \\ \begin{array}{cc} \uparrow & \uparrow \\ x & y \end{array} \end{array}$$



# RAPID FIRE ROUND



Find inverse of following functions :

(iii)  $f: [1, \infty) \rightarrow [1, \infty), f(x) = 2^{x(x-1)},$

[JEE-1999, 2M]

$$\# y = 2^{x^2-x}$$

$$x \leftrightarrow y$$

$$x = 2^{y^2-y}$$

$$\log_2 x = \log_2 2^{y^2-y}$$

$$\log_2 x = (y^2 - y) \log_2 2$$

$$\# y^2 - y - \log_2 x = 0$$

$$y = \frac{+1 \pm \sqrt{1 + 4 \log_2 x}}{2}$$

$$f^{-1}(x) = y = \frac{1 + \sqrt{1 + 4 \log_2 x}}{2} \quad \checkmark$$

$$y = \frac{1 - \sqrt{1 + 4 \log_2 x}}{2} \quad \times$$

$$x=1, y=1 \quad (1,1) \rightarrow (1,1)$$

$$y = \frac{1+1}{2} = 1$$

$$y = \frac{1-1}{2} = 0$$

#  $f: [1, \infty) \rightarrow [1, \infty)$ ,  $f(x) = 2^{x^2 - x}$  ✓

#  $f^{-1}: [1, \infty) \rightarrow [1, \infty)$ ,  $f^{-1}(x) = \frac{1 + \sqrt{1 + 4 \log_2 x}}{2}$  ✓

↙ ↘

↻ 1-1, Onto



## Question [JEE Mains-2020]

76%

The inverse function of  $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$ ,  $x \in (-1, 1)$  is

- A  $\frac{1}{4} \log_e \left( \frac{1+x}{1-x} \right)$
- B  $\frac{1}{4} \log_e \left( \frac{1-x}{1+x} \right)$
- ☒ C  $\frac{\log_8 e}{4} \log_e \left( \frac{1+x}{1-x} \right)$
- D  $\frac{\log_8 e}{4} \log_e \left( \frac{1-x}{1+x} \right)$

Handwritten solution:

$$y = \frac{8^{2x} - \frac{1}{8^{2x}}}{8^{2x} + \frac{1}{8^{2x}}} = \frac{8^{4x} - 1}{8^{4x} + 1}$$

$\cdot C + d$

$x = \frac{8^{4y} - 1}{8^{4y} + 1}$

$x \leftrightarrow y$

$$x + 1 = 8^{4y} - x + x + 1 = 8^{4y}$$
$$\frac{1+x}{1-x} = 8^{4y}$$
$$\frac{\log_8 e}{4} \ln \left( \frac{1+x}{1-x} \right) = y$$
$$x 8^{4y} + x = 8^{4y} - 1$$



## PROPERTIES OF INVERSE FUNCTION



- ✓ **Property-1:** Only Bijective function are invertible.
- ✓ **Property-2:** Inverse of a **bijection** is also a **bijection** and is **unique**.
- ✓ **Property-3:** Inverse of composite function:

$$\left(f(g(x))\right)^{-1} = g^{-1}\left(f^{-1}(x)\right) = g^{-1} \circ f^{-1}(x)$$

$$\left(f(g(h(x)))\right)^{-1} = h^{-1}\left(g^{-1}\left(f^{-1}(x)\right)\right) = \left(h^{-1} \circ g^{-1} \circ f^{-1}\right)(x)$$





## PROPERTIES OF INVERSE FUNCTION

**Property -04:** \*  **$g(x)$  is inverse of  $f(x)$**

$$g = f^{-1} \text{ or } f = g^{-1}$$

\*  $g = f^{-1}$  or  $g^{-1} = f$

\*  $g(f(x)) = x$  or  $f(g(x)) = x$

**Also Note:**

If  $f: A \rightarrow B$  &  $f(x)$  is an invertible function, then  **$f(f^{-1}(x)) = x \forall x \in B$**  and  $f^{-1}(f(x)) = x \forall x \in A$  or  $f(f^{-1}(x)) = I_B$  &  $f^{-1}(f(x)) = I_A$

Identity func<sup>n</sup>.  
on set B.

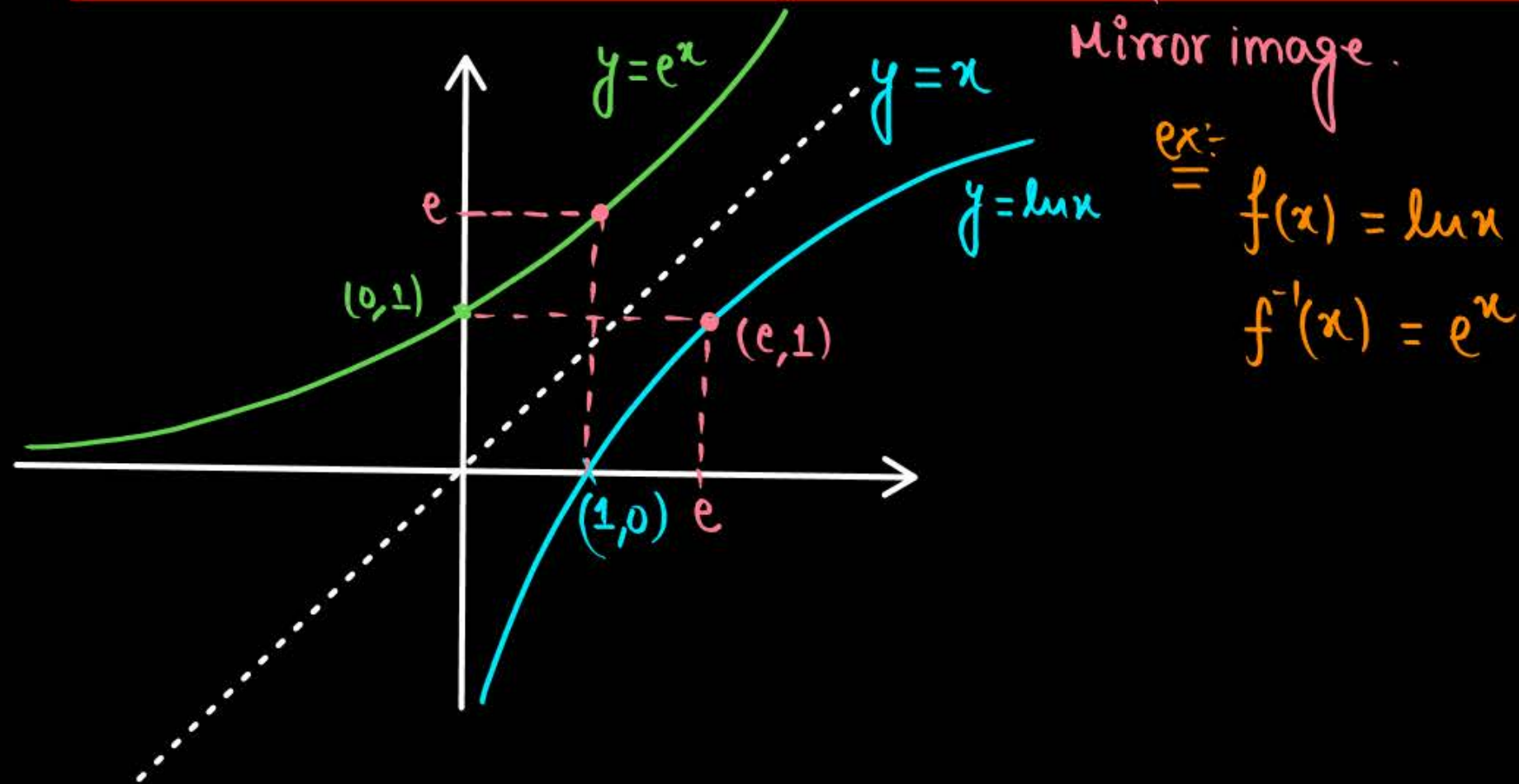




# PROPERTIES OF INVERSE FUNCTION

**Property -05:** If  $(\alpha, \beta)$  lies on graph of  $f(x)$ , then  $(\beta, \alpha)$  lies on graph of  $f^{-1}(x)$

$\Rightarrow$  Graph of  $f(x)$  &  $f^{-1}(x)$  is **symmetrical** about **line  $y = x$** .





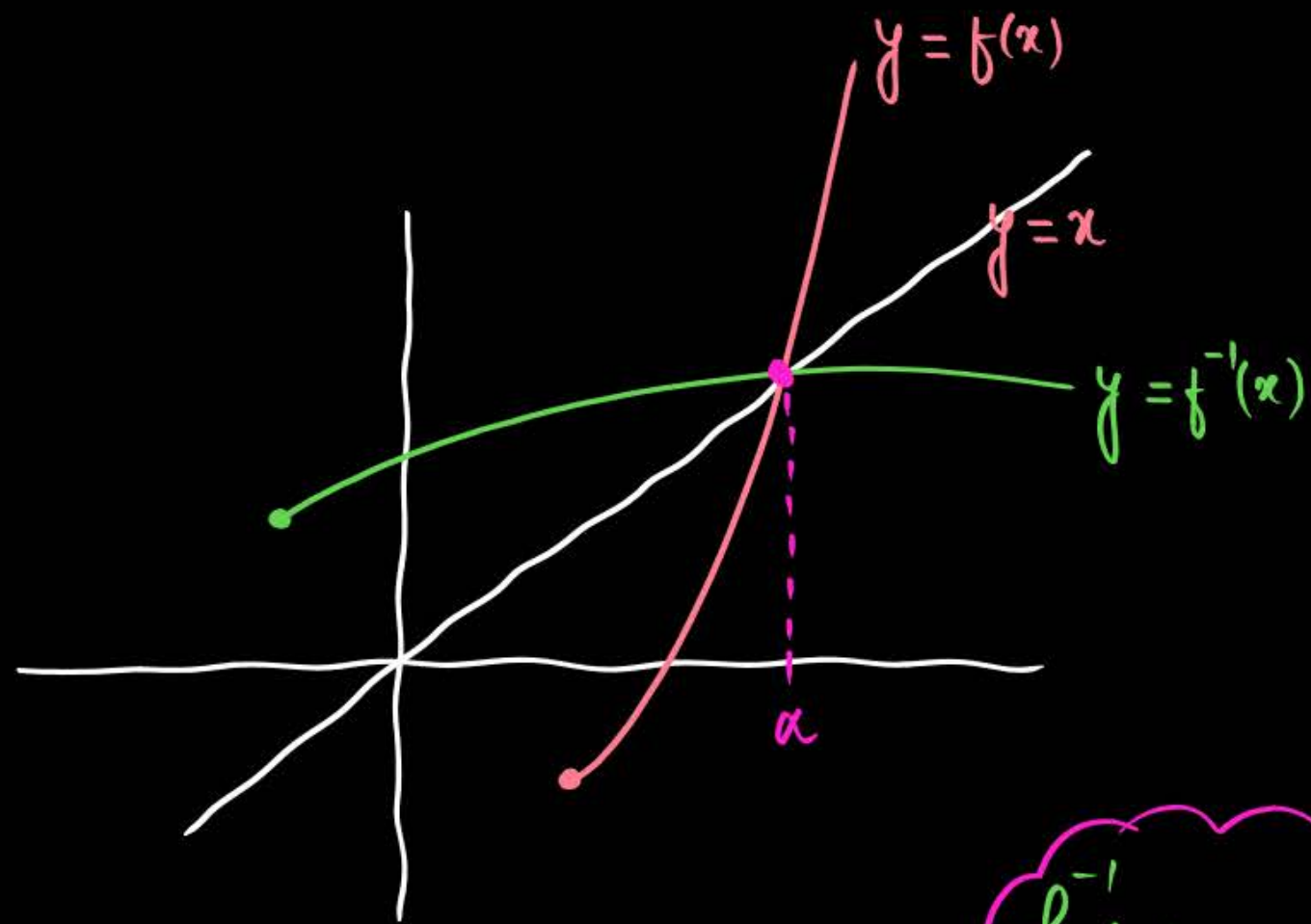
## PROPERTIES OF INVERSE FUNCTION

**Property -06:** Solution of equation  $f(x) = f^{-1}(x)$  are exactly same as that of  $f(x) = x$ , provided  $f(x)$  &  $f^{-1}(x)$  do not intersect outside of line  $y = x$ .

**Note:**

If  $f(x)$  &  $f^{-1}(x)$  intersects at exactly two point at A & B, outside of the line  $y = x$ , then slope of line AB is -1.

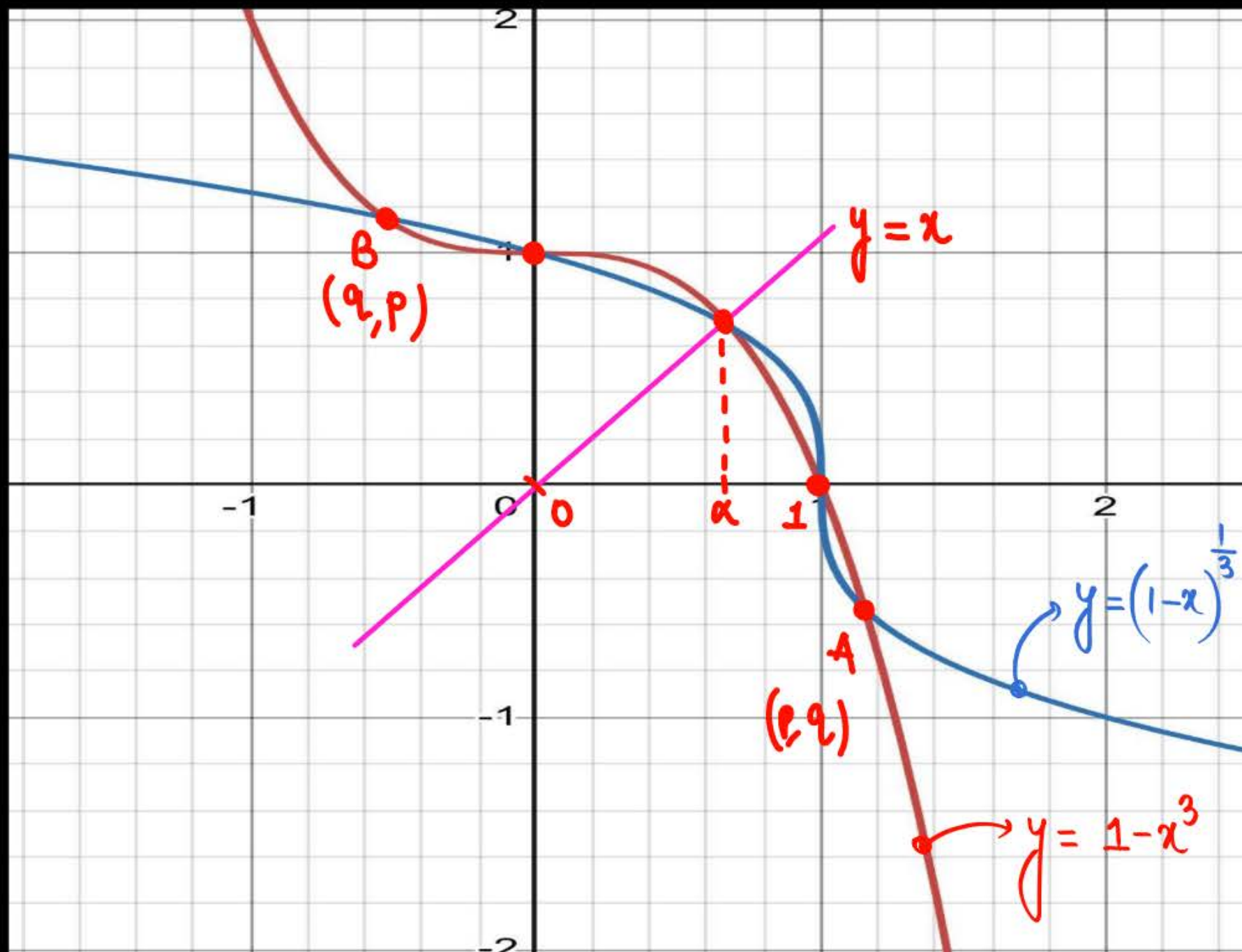




$$\# \quad f^{-1}(x) = f(x) \rightarrow \text{sol.}^n \Rightarrow x = \alpha$$

OR

$$x = f(x) \quad \text{OR} \quad f^{-1}(x) = x$$



$$f(x) = f^{-1}(x)$$

$$\# \quad (1-x^3) = (1-x)^{\frac{1}{3}}$$

$\rightarrow \underline{0}, \underline{1}, \underline{\alpha}, \underline{\beta}, \underline{\gamma}.$

$$\# \quad 1-x^3 = x$$

$\rightarrow \alpha$

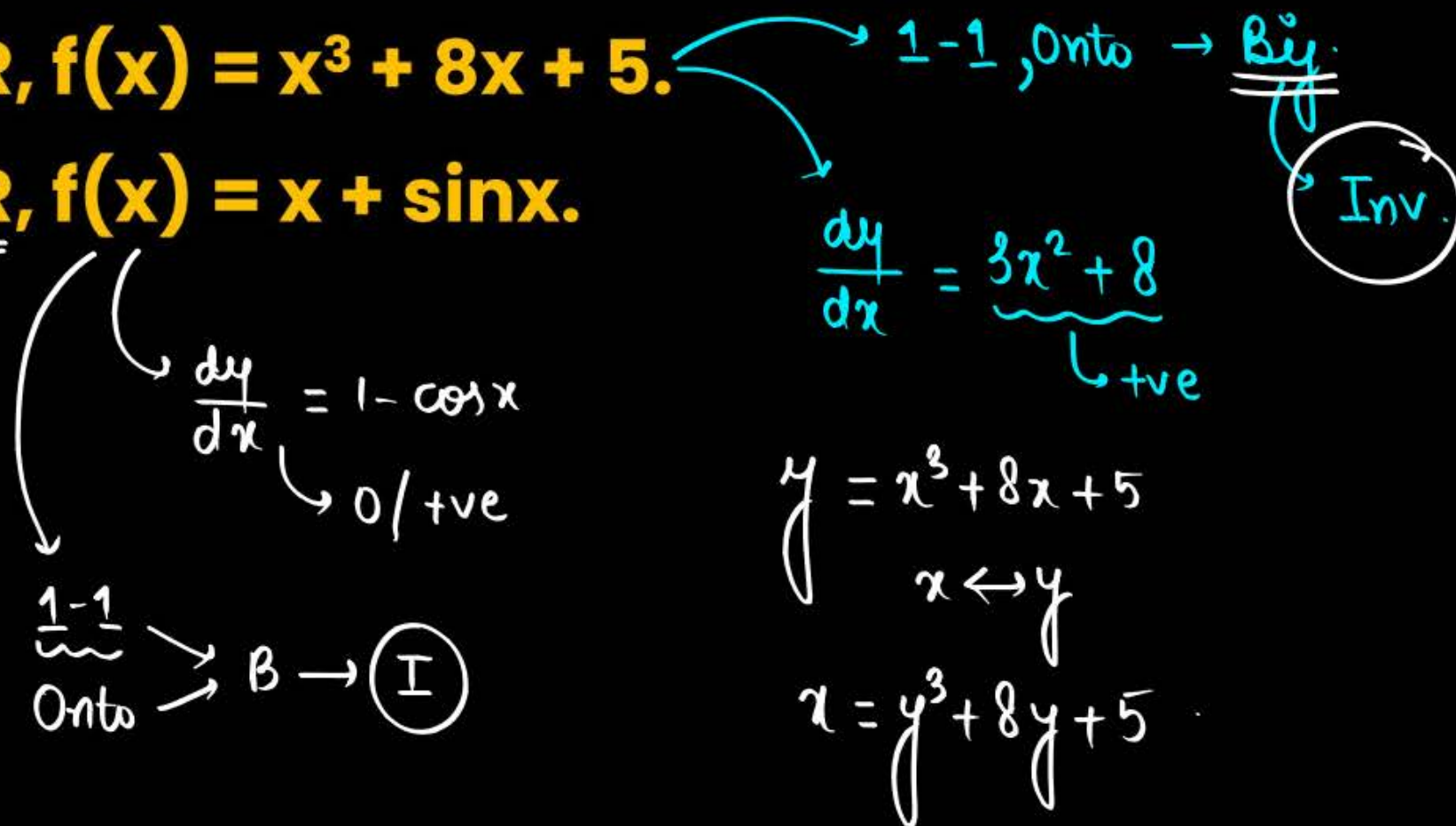


## # Note -01:

There are functions which are **invertible** but we cannot get them in explicit form ('y' ko akela nhi kiya ja skta)

Ex.:  $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 + 8x + 5.$

Ex.:  $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x + \sin x.$



## Acidic Question

Solve:

(i)  $f(x) = f^{-1}(x)$ ,  $x \in [-2\pi, 2\pi]$  where  $f(x) = x + \sin x$ .

(ii)  $x(x-1) = \frac{1+\sqrt{1+4x}}{2}$ ,  $x \in [\frac{1}{2}, \infty)$

$\underline{M-I}:$   
 $2(x^2 - x) - 1 = \sqrt{1+4x}$

# SBS

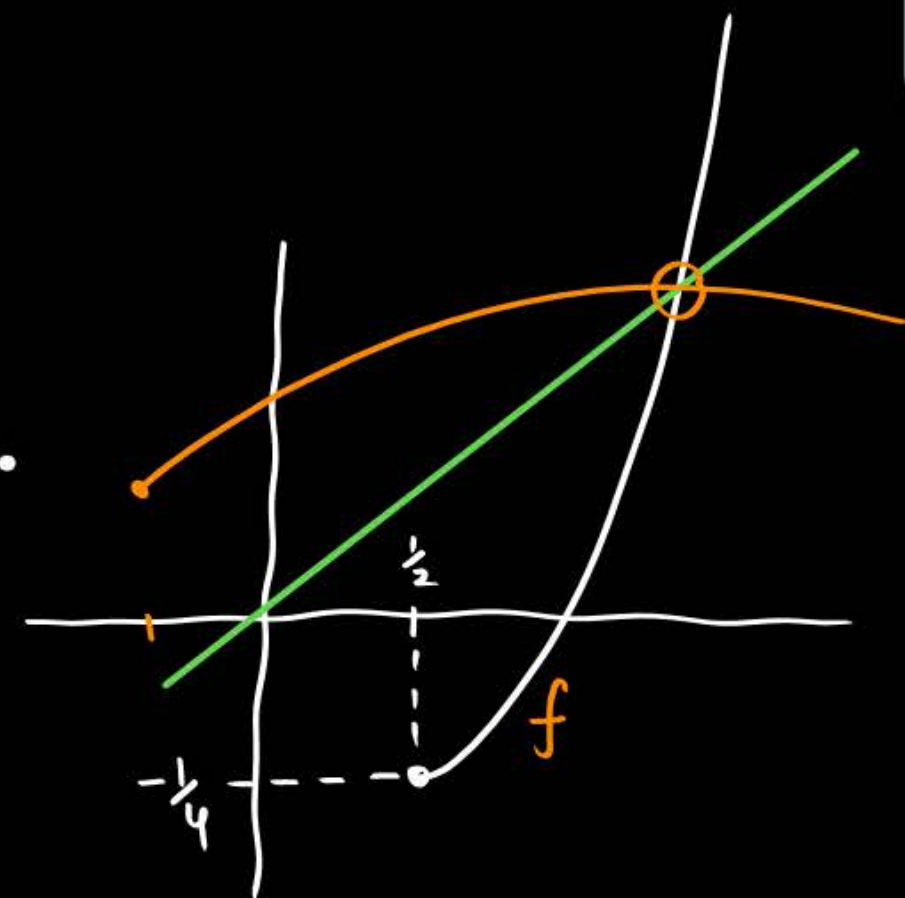


$\underline{M-II}:$   $f(x) = f^{-1}(x)$

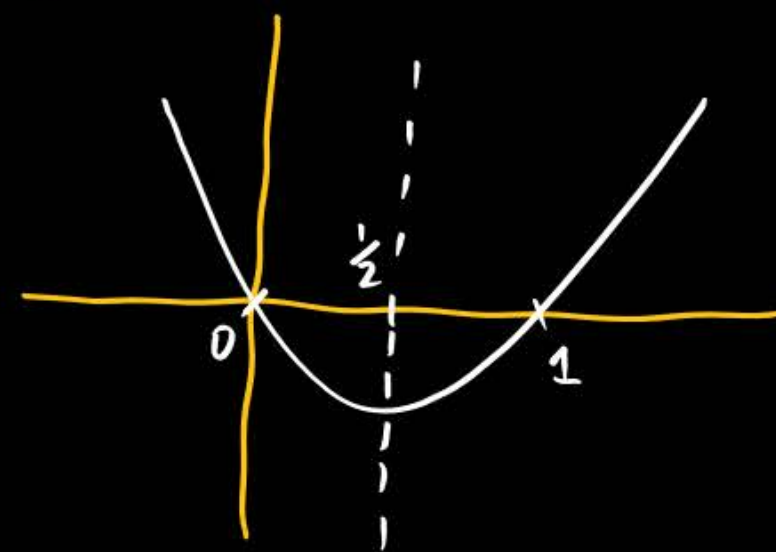
$f(x) = x$

$x^2 - x = x$

$x^2 - 2x = 0 \rightarrow x = 0, 2$



$f(x) = x(x-1) = x^2 - x$



## Acidic Question



Solve:

(i)  $f(x) = f^{-1}(x)$ ,  $x \in [-2\pi, 2\pi]$  where  $f(x) = x - \sin x$ .

(ii)  $x(x-1) = \frac{1+\sqrt{1+4x}}{2}$ ,  $x \in \left[\frac{1}{2}, \infty\right)$

$$\begin{aligned}\frac{dy}{dx} &= 1 - \cos x = 0 \\ &\hookrightarrow \cos x = 1 \\ &\downarrow \\ &-2\pi, 0, 2\pi\end{aligned}$$

$$\frac{dy}{dx} = 1 - \cos x$$

0 / +ve

$$x^2 - x$$



$$y = x - \sin x$$

$$y = x$$

$$\frac{dy}{dx} = 1 - \cos x = 0$$

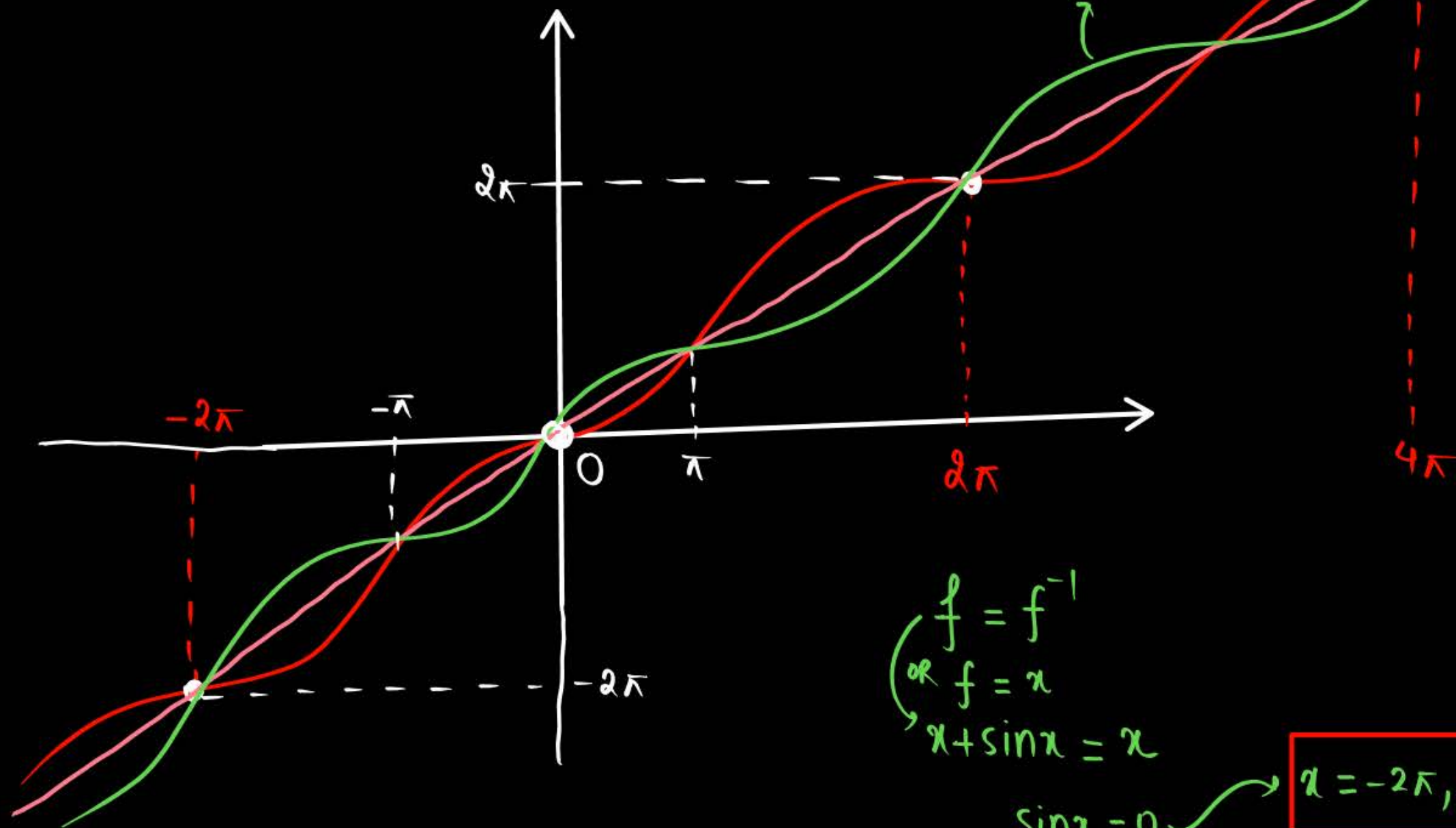
$$\rightarrow \cos x = 1$$

$$\downarrow$$

$$-2\pi, 0, 2\pi$$

$$\frac{dy}{dx} = 1 - \cos x$$

$$\rightarrow 0 \mid \text{+ve}$$



$$f = f^{-1}$$

$$\text{or } f = x$$

$$x + \sin x = x$$

$$\sin x = 0$$

$$x = -2\pi, -\pi, 0, \pi, 2\pi$$

## IMPORTANT NOTE



### # Note -02:

If asked value of  $f^{-1}(\alpha)$  that means solve  $f(x) = \alpha$  and value of 'x' is our answer.

Ex. :  $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 + 8x + 5$ , then  $f^{-1}(-4) = \underline{\underline{-1}}$ .

Ex. : If  $f(x) = \frac{3}{2} + \sqrt{x - \frac{7}{4}}$ , then  $f^{-1} \circ f(8) = 8$ .

$\downarrow$   
 $\underline{f^{-1}(f(8))}$

$$\begin{aligned} \# \quad f^{-1}(-4) &= x \rightarrow \textcircled{-1} \\ -4 &= f(x) \\ -4 &= x^3 + 8x + 5 \\ \downarrow \\ x &= -1 \end{aligned}$$

## IMPORTANT NOTE



### # Note -03 : Important Observations

Exponential Function  $\xrightleftharpoons{\text{Inverse}}$  log

Function with power 'n'  $\xrightleftharpoons{\text{Inverse}}$  Function with power  $\frac{1}{n}$   
 $\hookrightarrow x^n$   $\hookrightarrow x^{\frac{1}{n}}$

Ex. :  $f(x) = x^2 + x + 1$   $\xrightleftharpoons{\text{Inverse}}$   $g(x) = \frac{\sqrt{4x-3}-1}{2}$

Ex. :  $f(x) = \log_2(x^3 + 1)$   $\xrightleftharpoons{\text{Inverse}}$   $g(x) = (2^x - 1)^{1/3}$



**DO IT BY YOURSELF (DIBY)**

## # DIBY-186

Let  $A = \{1, 2, 3, \dots, 10\}$  and  $f: A \rightarrow A$  be defined as  $f(k) = \begin{cases} k+1, & \text{if } k \text{ is odd} \\ k, & \text{if } k \text{ is even} \end{cases}$ . Then the number of possible functions  $g: A \rightarrow A$  such that  $g \circ f = f$  is **[JEE Mains-2021]**

(A)  $10^5$       (B)  ${}^{10}C_5$       (C)  $5^5$       (D)  $5!$

[Ans. : (A)]

## # DIBY-187

Let  $f$  and  $g$  be functions given by

$$f(x) = -1 + |x - 1|, \quad -1 \leq x \leq 3 \quad \text{and} \quad g(x) = 2 - |x + 1|, \quad -2 \leq x \leq 2$$

Find an explicit representation of  $f \circ g$  and  $g \circ f$ .

$$[\text{Ans. : } (f \circ g)x = \begin{cases} x+1, & -2 \leq x < -1 \\ -x-1, & -1 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}$$

$$(g \circ f)x = \begin{cases} 1+x, & -1 \leq x < 1 \\ 3-x, & 1 \leq x \leq 3 \end{cases}]$$

## # DIBY-188

Let  $f : A \rightarrow B$  and  $g : B \rightarrow C$  be functions and  $g \circ f : A \rightarrow C$ .

Which of the following statements is true?

- (A) If  $g \circ f$  is one-one, then  $f$  and  $g$  both are one-one.
- (B) If  $g \circ f$  is one-one, then  $f$  is one-one.
- (C) If  $g \circ f$  is a bijection, then  $f$  is one-one and  $g$  is onto.
- (D) If  $f$  and  $g$  are both one-one, then  $g \circ f$  is one-one.

[Ans. : B, C, D]

## # DIBY-189

Let  $f_1(x) = x$ ,  $f_2(x) = 1 - x$ ;  $f_3(x) = \frac{1}{x}$ ,  $f_4(x) = \frac{1}{1-x}$ ;  $f_5(x) = \frac{x}{x-1}$ ;  $f_6(x) = \frac{x-1}{x}$ . Suppose that  $f_6(f_m(x)) = f_4(x)$  and  $f_n(f_4(x)) = f_3(x)$ , then find the value of  $m$  and  $n$ .

[Ans. :  $m = 6, n = 5$ ]





# Homework



**# Re-attempt all good Questions of Class**

**# DIBY**

*It is never about end result  
It's always about Journey*



**THANK**  
*You*

**#futureellTians**